STEG STRUCTURAL TRANSFORMATION AND ECONOMIC GROWTH

Small Research Grant Proposal

Project Title:
Spatial Structural Change with Financial Frictions
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1. INTRODUCTION

Standard trade theory suggests a positive agricultural price shock should increase the labour supplied to agriculture. However, recent literature (Laskievic (2021)) and our own global panel analysis document the opposite in many emerging economies: global agricultural booms accelerate labour *exit*. We resolve this puzzle using a heterogeneous agent model with financial frictions and spatial migration costs. A large empirical literature documents that liquidity constraints trap subsistence farmers in unproductive agriculture (such as Bryan et al. (2014), and Angelucci (2015)). Motivated by this literature, we posit that macroeconomic price shocks can generate windfalls, allowing farmers to pay fixed migration costs. This resolves the apparent puzzle that *positive* macroeconomic shocks lead to *exit*, and identifies a novel channel connecting trade shocks to structural transformation.

We will use a combination of reduced-form methods and a structural model to quantify the importance of this channel. First, using granular IPUMS census data from **Kenya** and FAO-GAEZ geospatial data, we will construct a Bartik shift-share design that interacts global agricultural price shocks (from World Bank Pink Sheets) with exogenous crop suitability to identify the causal effect of price shocks on migration. This design follows existing work on Brazil by Laskievic (2021). Using IPUMS and FAO-GAEZ at a higher resolution allows us to study these spatial dynamics at the county level.

Secondly, we will build a heterogeneous agent development model calibrated granularly to Kenya. This model will be solved using the **Sequence Space Jacobian (SSJ)** method from Auclert et al. (2021). Bringing the SSJ to a dynamic spatial setting with trade represents a new methodological step. The SSJ allows for applications with incomplete

markets, savings and heterogeneous agents which is not possible using the "Dynamic Hat Algebra" approach from Caliendo et al. (2019) which is current frontier in most of the dynamic spatial literature.

Our theoretical framework allows us to measure the strength of financial frictions in slowing down structural change in Kenya by inhibiting farmers from moving to urban centres. Crucially, we are able to do this in a full global general equilibrium trade model. Our model allows us to simulate two policy-relevant counterfactuals:
1) Did the China-driven commodity supercycle in the 2000s inadvertently speed up Kenya's structural change?
2) Do EU common agricultural policy subsidies, by depressing global prices, actively trap Kenyan farmers in subsistence agriculture?

Project feasibility is supported by our preliminary work: we have harmonized trade data, developed the prototype model, and solved the domestic heterogeneous agent block. We have also documented the key negative agricultural employment response in emerging markets, including Kenya. We seek funding to buy out one quarter of teaching obligations, doubling our research time to complete the granular quantification.

2. METHODOLOGY

1. **Reduced Form:** We will establish the causal link between agricultural price shocks and structural transformation using Jorda's (2005) Local Projections.

   i) **Global Block:** We use a panel of 45 developing countries (GGDC ETD). To address endogeneity, we construct a Bartik instrument. The *shift* is the change in real global commodity prices (World Bank GEM, deflated by MUV). The *share* is the "Predicted Agricultural Share" constructed using FAO-GAEZ v4 potential yields. By using potential yields rather than observed production, we isolate variation driven by exogenous endowment rather than endogenous crop choice.

   ii) **Granular Block (Kenya):** We zoom in on Kenya using IPUMS International census microdata (1969, 1979, 1989, 1999, 2009, 2019). We will exploit sub-national variation in crop suitability and global agricultural price shocks (deflated by manufacturing prices) to test our hypothesis: that migration rates increase after specific households receive positive price shocks that push their savings high enough to allow migration.

2. **Structural Model:** We extend the dynamic spatial trade framework of Caliendo et al. (2019) by introducing heterogeneous agents with incomplete markets in the spirit of Aiyagari (1994).

Model Features: Agents face idiosyncratic productivity shocks and a borrowing constraint. Aggregate fundamentals vary over time but agents have perfect foresight over aggregates. Migration is a dynamic choice with a fixed cost. The model can generate a poverty trap where agents with low wealth do not migrate unless they get either high idiosyncratic realizations of productivity or find themselves in an aggregate state of high fundamentals ("boom"). This heterogeneous agent structure is then embedded in a very simple global trade model (Eaton-Kortum with a three sector input-output network). Embedding the spatial model inside of a global trade model allows us to model trade-shock counterfactuals in an internally consistent framework without ad-hoc assumptions.

Solution Method: We solve for the perfect foresight transition path using the **Sequence Space Jacobian (SSJ)** method. Unlike standard "dynamic hat algebra" which is limited to representative agent models and hand-to-mouth households, the SSJ allows us to compute the full general equilibrium non-linear transition path in models that feature incomplete markets and heterogeneous households. Extending the SSJ framework to the spatial setting with discrete choice migration decisions inside of a quantitive trade model is the main technical innovation of the project.

3. POLICY RELEVANCE AND COUNTRY FOCUS

Primary Focus: Kenya (Sub-Saharan Africa). We focus on Kenya due to its rich IPUMS census data, heavy export reliance on crops, and large regional disparities. Furthermore, our shift-share analysis confirms that Kenya is a significant negative responder to global agricultural price shocks: farmers leave agriculture when prices rise.

Policy Implications: Both the reduced form analysis and structural model counterfactuals are designed with direct policy relevance in mind.

1) **EU Common Agricultural Policy:** We will simulate a counterfactual global equilibrium without EU Common Agricultural Policy (CAP) subsidies. We conjecture that CAP subsidies, by depressing global prices, reduce farmer savings and delay migration. Our model will measure whether advanced economy protectionism drives agricultural poverty traps in SSA.
2) **The China Shock:** We will simulate the China-driven commodity supercycle (2000-2014). The China-driven agricultural boom may have accelerated Kenyan urbanization by relaxing rural liquidity constraints.
3) **Actionable Policy:** Our results build on literature linking rural credit to structural transformation. We highlight trade as a novel mechanism affecting credit conditions and mobility. This understanding is relevant for policymakers in SSA and developed economies, altering the implications of export subsidies and tariffs. We are motivated disseminate findings via policy briefs to Kenyan policymakers or VoxEU if possible.

4. RELEVANCE TO STEG RESEARCH AGENDA, RESEARCH THEMES, AND CROSS-CUTTING ISSUES

This project directly addresses two STEG research themes:

Theme 3: Agricultural Productivity and Sectoral Gaps. This project was initially motivated by our observation that many macroeconomic structural change models do not get the agricultural productivity gap right. We add to the micro-founded theories regarding the persistence of the agricultural productivity gap by modelling the interaction of spatial and financial frictions inside of a general equilibrium model with structural change. In our model, small spatial wage differentials due to migration costs are increased by financial frictions that inhibit paying these costs.

Theme 4: Trade and Spatial Frictions. We explicitly model how international terms-of-trade shocks interact with internal spatial mobility costs to determine the pace of structural change. Our main counterfactual policy exercises will look at the effect of trade shocks on migration out of agriculture. Furthermore, by using the SSJ method to analyze

transition paths, we are able to look at inequality dynamics, tracking how trade affects the wealth distribution during the development process.

5. DETAILED WORKPLAN

Feasibility: We have made the following key strides:

At low granularity we have documented the key fact in reduced form: postive agricultural price shocks are associated with migration *out* of agriculture for many emerging economies.

The Eaton-Kortum Input-Output trade block is already close to being finished. With full gross-sector level prices processed and model inversion for TFP and iceberg trade costs done.

The heterogeneous agent spatial prototype "domestic block" is already solved with the SSJ and we can generate the key stylized facts discussed above.

The remaining work focuses on the granular quantification. We will process the IPUMS data to estimate the Kenyan model parameters and establish granular reduced-form facts regarding migration heterogeneity (by region and income). Finally, we will align the model and empirical impulse responses. Our prototype results demonstrate the computational feasibility of this approach.

6. EXPECTED OUTPUTS

Academic Paper: "Spatial Structural Change and Trade with Financial Frictions". We are targeting a top macro or development field journal (e.g. AEJ macro, JDE) or depending on later-stage feedback possibly a top general interest journal.

Public Dataset: A harmonized dataset of "Agricultural Price Exposure" indices for 45 countries and Kenyan counties, available for other researchers.

Policy Brief: A summary analyzing the impact of international policies on developing economy labour mobility - with a detailed focus on Kenya.

7. CONTRIBUTIONS TO EXPANDING THE FIELD

This project expands the study of trade, macro-development, and spatial economics in a few ways

1) We will document the extent to which macroeconomic shocks can indirectly work like the micro-credit loans commonly studied in the empirical development literature. We will measure this channel in both reduced form and study implication in our model.

2) **Methodologically:** We extend the Sequence Space Jacobian method to migration problems in economies with structural change. We demonstrate that this allows for the study of question of savings, inequality and incomplete markets in a way that is not possible with existing "Dynamic Hat Algebra" methods in dynamic spatial models.

9. OTHER FUNDING

We have no other current funding for this specific project, or others.

10. ETHICS

Yes, we will comply with the Code of Conduct.

Human Subjects: No. This project relies exclusively on secondary, anonymized public data (IPUMS, GGDC, World Bank). No primary data collection is involved.

11. PRIVACY

Does the research involve personal data? **No.** All microdata (IPUMS) is anonymized.

12. INTELLECTUAL PROPERTY RIGHTS

Yes. We agree to the IPR policy.

13. OPEN AND ENHANCED ACCESS TO RESEARCH OUTPUTS

We confirm compliance with the Open Access Policy. All code and derived datasets will be hosted on a public GitHub repository upon publication.

14. LIST OF REFERENCES CITED

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